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2+3+5 = 10m

DIFFERENTIAL EQUATION

Definition: DE: An equation involving derivatives of the dependent variable w.r.t independent variable is called a DE.

Eg: $\frac{d^3y}{dx^3} - 2y \frac{dy}{dx} + 7y = 0$

$2y \frac{dy}{dx} - y = 0$

There are 2 types of DE's:

(a) Partial differential equation (PDE):

DE's involving derivatives of the dependent variable w.r.t more than one independent variables, are called PDEs.

Eg: $\frac{\partial p}{\partial x} + \frac{\partial p}{\partial y} + \frac{\partial p}{\partial z} = 7$

(b) Ordinary differential equation (ODE):

DE's involving derivatives of the dependent variable w.r.t to only one independent variable, are called ODE's.

Order and degree of DE:

(a) Order: Order of a DE is defined as the order of highest order derivative involved in the given DE.

(b) Degree: Degree of DE is the highest power of the highest order derivative in the given DE.

NOTE: Order and degree should be found only in the absence of radical powers (powers should be only in whole no.s).

** If the derivative is present, as $\cos\left(\frac{dy}{dx}\right)$ or $e^{\frac{dy}{dx}}$ or $\log\left(\frac{dy}{dx}\right)$, then degree is not defined. ($\because$ such derivatives do not form a polynomial in y')

EXERCISE 9.1

Determine order and degree:

1) $\frac{d^4y}{dx^4} + \sin(y''') = 0$

Order (O) - 4

Degree (D) - not defined

2) $(y')' + sy = 0$ O - 1, D - 1

3) $\left(\frac{ds}{dt}\right)^4 + 3s\frac{d^2s}{dt^2} = 0$ O - 2, D - 1

4) $\left(\frac{d^2y}{dx^2}\right)^2 + \cos\left(\frac{dy}{dx}\right) = 0$ O - 2, D - not defined

5) $\frac{d^2y}{dx^2} = \cos 3x + \sin 3x$ O - 2, D - 1

6) $(y''')^2 + (y'')^3 + (y')^4 + y^5 = 0$ O - 3, D - 2

7) $y''' + 2y'' + y' = 0$ O - 3, D - 1

8) $y' + y = e^x$ O - 1, D - 1

9) $y'' + (y')^2 + 2y = 0$ O - 2, D - 1

10) $y'' + 2y' + \sin y = 0$ O - 2, D - 1

11) $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^2 + \sin\left(\frac{dy}{dx}\right) + 1 = 0$ O - 2, D - not defined

12) $2x^2\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + y = 0$ O - 2, D - 1

Eg1] (a) $\frac{dy}{dx} - \cos x = 0$ $0 - 1, D = 1$ $\text{pol} = yx$ [r

(b) $xy \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx}\right)^2 - y \frac{dy}{dx} = 0$ $0 - 2, D = 1$

(c) $y''' + y^2 e^y = 0$ $0 - 3, D = \text{not defined}$

GS and PS of a DE:

GS - General solution ; PE - Particular solution

- * GS is the solution consisting of the constant c/c and c .
- * Using initial conditions, substituting for c and c , in GS, we get PS

EXERCISE 9.2

In the following problems, verify that the given functions is a solution of the corresponding DE.

1) $y = e^x + 1$: $y'' - y' = 0$

$y' = e^x$

$y'' = e^x$

RHS = $y'' - y' = e^x - e^x = 0 = \text{RHS}$

Hence verified.

2) $y = x^2 + 2x + C$: $y' - 2x - 2 = 0$

$y' = 2x + 2$

$y' - 2x - 2 = 0$

Hence verified

3) $y = \cos x + C$: $y' + \sin x = 0$

$y' = -\sin x$

$y' + \sin x = 0$ Hence verified

$$7] \quad xy = \log y + C \quad \therefore y' = \frac{y^2}{1-xy}$$

$$xy' + y = \frac{1}{y} \cdot y' \quad 0 = \frac{1}{y} y' - \frac{1}{y^2} \cdot y^2$$

$$y' \left(\frac{1}{y} - x \right) = 0 \quad \therefore \frac{1}{y} - x = 0 \quad \therefore \frac{1}{y} = x \quad \therefore y = \frac{1}{x}$$

$$y' \left(\frac{1-xy}{y} \right) = 0 \quad \therefore y' = \frac{y^2}{1-xy} \quad \text{Hence verified 29 two 29}$$

$$9] \quad x + y = \tan^{-1} y \quad \therefore y^2 y' + y^2 + 1 = 0$$

$$1 + y^2 = \frac{1}{1+y^2} \cdot y'$$

$$1 = \left(\frac{1}{1+y^2} - 1 \right) y'$$

$$1 + y^2 = -y^2 y'$$

$$y^2 y' + y^2 + 1 = 0$$

** 11] The no. of arbitrary constants in the GS of a DE of 4th order are:

- (a) 0 (c) 3
(b) 2 (d) 4

12] The no. of arbitrary constants in PS of a DE of 3rd order are:

- (a) 3 (c) 1
(b) 2 (d) 0

$$\begin{aligned}
 8) \quad y - \cos y &= x \quad : (y \sin y + \cos y + x) y' = y = p \quad [1] \\
 y' + \sin y y' &= 1 \\
 y' [\sin y + 1] &= 1 \\
 \text{Multiply by } y \text{ throughout} &=, [y = x + \cos y] = p \\
 y y' (\sin y + 1) &= y \\
 y' (x + \cos y) (\sin y + 1) &= y \\
 y' [x \sin y + x + \cos y \sin y + \cos y] &= y \quad [2] \\
 y' [(y - \cos y) \sin y + x + \cos y \sin y + \cos y] &= y \\
 y' [y \sin y - \cos y \sin y + x + \cos y \sin y + \cos y] &= y \\
 y' [y \sin y + \cos y + x] &= y
 \end{aligned}$$

$$\begin{aligned}
 10) \quad y &= \sqrt{a^2 - x^2} \quad : x + y \frac{dy}{dx} = 0 \quad [1] \\
 \frac{dy}{dx} &= \frac{-2x}{2\sqrt{a^2 - x^2}} \\
 \frac{dy}{dx} &= \frac{-x}{\sqrt{a^2 - x^2}} \\
 \text{Multiply by } y \text{ on both sides, } x + y \frac{dy}{dx} &= 0 \\
 y \frac{dy}{dx} &= \frac{-x}{\sqrt{a^2 - x^2}} \sqrt{a^2 - x^2} \\
 x + y \frac{dy}{dx} &= 0
 \end{aligned}$$

(3m) Formation of DE :

To eliminate one constant, differentiate once.

To eliminate two constants, differentiate twice.

EXERCISE 9.3

$$\begin{aligned}
 1) \quad \frac{x}{a} + \frac{y}{b} &= 1 \\
 \frac{1}{a} + \frac{1}{b} y' &= 0 \\
 y' &= -\frac{b}{a} \\
 y'' &= 0
 \end{aligned}$$

$$2] \quad y^2 = a(b^2 - x^2)$$

$$2xyy' = a(-2x) \quad 0 = \left(\frac{2x}{b} \right) \frac{y}{x} + \frac{y}{x} \frac{y}{b} + \frac{y}{x} \frac{y}{b}$$

$$\frac{yy'}{x} = -a$$

$$\frac{x(yy'' + (y')^2) - yy'}{x^2} = 0$$

$$xyy'' + x(y')^2 - yy' = 0$$

** 6] Form DE of the family of circles such + touching y-axis at origin.

NOTE: General circle: $x^2 + y^2 + 2gx + 2fy + c = 0$; center $= (-g, -f)$

- circle passes through origin $\Rightarrow c = 0$
- centre on x-axis $\Rightarrow f = 0$
- centre on y-axis $\Rightarrow g = 0$

Sol] Coaxial system:

$$\text{general eq: } x^2 + y^2 + 2gx + 2fy + c = 0$$

center on x-axis $\Rightarrow f = 0$, passing through $(0,0) \Rightarrow c = 0$

$$\therefore \text{Req circle: } x^2 + y^2 + 2gx + 0 = 0 \rightarrow (1)$$

$$2x + 2y \frac{dy}{dx} + 2g = 0 \rightarrow (2)$$

Using (2) in (1)

$$x^2 + y^2 + (-2x - 2y \frac{dy}{dx})x = 0$$

$$y^2 - x^2 - 2xy \frac{dy}{dx} = 0$$

** Eg 1] Form DE of family of circles touching x-axis at origin.

Sol] Passes through $(0,0) \Rightarrow c = 0$

Centre on y-axis $\Rightarrow g = 0$

$$\therefore \text{family of } 0^{th}: x^2 + y^2 + 2fy = 0 \rightarrow (1)$$

$$2x + 2y \frac{dy}{dx} + 2f \frac{dy}{dx} = 0 \rightarrow (2)$$

Using ① in ②:

$$2x + 2y \frac{dy}{dx} + \frac{dy}{dx} \left(\frac{-x^2 - y^2}{y} \right) = 0$$

multiply by 'y'

$$2xy + 2y^2 \frac{dy}{dx} - x^2 \frac{dy}{dx} - y^2 \frac{dy}{dx} = 0$$

$$2xy + (y^2 - x^2) \frac{dy}{dx} = 0$$

** Eg 6] Form DE of family of ellipses having foci on x-axis and centre at origin.

Sol] Gn: Family of ellipses:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (a > b)$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{2x}{a^2} \cdot \frac{b^2}{2y}$$

$$\frac{yy'}{x} = -\frac{b^2}{a^2}$$

$$\frac{x(yy'' + (y')^2) - yy'}{x^2} = 0$$

$$xyy'' + x(y')^2 - yy' = 0$$

$$xy \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 - y \frac{dy}{dx} = 0$$

7] Form DE of family of parabolas having vertex at origin and axis along +ve y-axis.

Sol] Gn: Family of parabolas

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$x^2 = 4ay \rightarrow (1)$

$2x = 4a \frac{dy}{dx}$

$2x = \frac{x^2}{y} \frac{dy}{dx}$

$2xy - x^2 \frac{dy}{dx} = 0$

8] Form DE of family of ellipses having foci on y-axis and centre at origin.

Sol.] Gn: family of ellipses

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (a < b)$

$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\frac{2x}{a^2} \cdot \frac{b^2}{2y}$

$\frac{yy_1}{x} = -\frac{b^2}{a^2}$

$\frac{x(yy_2 + y_1^2)}{x^2} - yy_1 = 0$

$xy \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 - y \frac{dy}{dx} = 0$

9] Form DE of family of hyperbolas having foci on x-axis and center at origin.

Sol.] Gn: family of hyperbola:

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$\frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0$

$\frac{dy}{dx} = \frac{2x}{a^2} \cdot \frac{b^2}{2y}$

$\frac{yy_1}{x} = \frac{b^2}{a^2}$

$$\frac{x(y_1 y_2 + y_1^2) - y y_1}{x^2} = 0$$

$$x y \frac{d^2 y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 - y \frac{dy}{dx} = 0$$

10) Form DE of family of circles having center on y-axis and radius 3 units.

Sol] Centre on y-axis = (0, k), r = 3

$$\text{Eq: } (x-h)^2 + (y-k)^2 = r^2$$

$$x^2 + (y-k)^2 = 9 \Rightarrow y-k = \sqrt{9-x^2}$$

$$x^2 + y^2 + k^2 - 2yk - 9 = 0$$

$$2x + 2y \frac{dy}{dx} - 2k \frac{dy}{dx} = 0$$

$$x + y \frac{dy}{dx} - k \frac{dy}{dx} = 0$$

$$x + \frac{dy}{dx}(y-k) = 0$$

$$x + \frac{dy}{dx}(\sqrt{9-x^2}) = 0$$

$$\left(\frac{dy}{dx} \right)^2 (9-x^2) = x^2$$

(or)

$$y - \sqrt{9-x^2} = k$$

$$\frac{dy}{dx} + \frac{x}{\sqrt{9-x^2}} = 0$$

DE's can be solved in many methods:

→ Variable separable form method:

- Here we separate the variables in given DE, integrate w.r.t corresponding variables and find G.S. Substituting initial values, get value of C and write P.S.

→ Homogeneous DE:

- If every term is of the same degree in given DE, then it is homogeneous DE.

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(a) If $\left(\frac{y}{x}\right)$ is dominative, then put $y = vx$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Now, DE reduces to variable separable form.

(b) If $\left(\frac{x}{y}\right)$ is dominative, then put $x = vy$

$$\frac{dx}{dy} = v + y \frac{dv}{dy}$$

(5m) $\rightarrow$ Linear DE (LDE) :

(a) Form: $\frac{dy}{dx} + Py = Q$

Integrating factor (IF) = $e^{\int P dx}$

GS : $y(IF) = \int Q(IF) dx$

(b) Form: $\frac{dx}{dy} + Px = Q$

IF : $e^{\int P dy}$

GS : $x(IF) = \int Q(IF) dy$

EXERCISE 9.4

1] $\frac{dy}{dx} = \frac{1 - \cos x}{1 + \cos x}$

$$\int dy = \int \frac{1 - \cos x}{1 + \cos x} dx$$

$$y = \int \frac{x \sin^2 \frac{x}{2}}{x \cos^2 \frac{x}{2}} dx = \int \tan^2 \left(\frac{x}{2}\right) dx = \int \sec^2 \frac{x}{2} - 1 dx$$

GS : $y = 2 \tan \frac{x}{2} - x + C$

2] $\frac{dy}{dx} = \sqrt{4 - y^2}$

$$\int \frac{dy}{\sqrt{4 - y^2}} = \int dx$$

$$\text{GS: } \sin^{-1}\left(\frac{y}{2}\right) = x + C$$

$$3] \frac{dy}{dx} + y = 1$$

$$\frac{dy}{dx} = 1 - y$$

$$\int \frac{dy}{1-y} = \int dx$$

$$\frac{\log |1-y|}{-1} = x + C$$

$$\text{GS: } \log \left| \frac{1}{1-y} \right| = x + C$$

$$4] \sec^2 x \tan y dx + \sec^2 y \tan x dy = 0$$

$$\sec^2 x \tan y dx = -\sec^2 y \tan x dy$$

$$\int \frac{\sec^2 x}{\tan x} dx = \int \frac{-\sec^2 y}{\tan y} dy$$

$$\log |\tan x| + \log |\tan y| = \log C$$

$$\log |\tan x \tan y| = \log C$$

$$\tan x \tan y = C$$

$$5] (e^x + e^{-x}) dy - (e^x - e^{-x}) dx = 0$$

$$(e^x + e^{-x}) dy = (e^x - e^{-x}) dx$$

$$\int dy = \int \frac{(e^x - e^{-x})}{(e^x + e^{-x})} dx$$

$$\text{GS: } y = \log(e^x + e^{-x}) + C$$

$$6] \frac{dy}{dx} = (1-x^2)(1+y^2)$$

$$\int \frac{dy}{1+y^2} = \int (1-x^2) dx$$

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GS: $\tan^{-1} y = x - \frac{x^3}{3} + C$

7] $y \log y \, dx - x \, dy = 0$
 $y \log y \, dx = x \, dy$
 $\int \frac{1}{x} \, dx = \int \frac{1}{y \log y} \, dy$
 $\log|x| = \log|\log y| + \log C \Rightarrow \log|x| = \log|C \log y|$
 $|x| = C \log y$

8] $x^5 \frac{dy}{dx} = -y^5$
 $\int \frac{dy}{y^5} = -\int \frac{dx}{x^5}$
 $\frac{y^{-4}}{-4} + \frac{x^{-4}}{-4} = C$
GS: $\frac{1}{y^4} + \frac{1}{x^4} = -4C = C$

9] $\frac{dy}{dx} = \sin^{-1} x$
 $\int dy = \int \sin^{-1} x \, dx$
 $y = \sin^{-1} x \cdot x - \int x \frac{1}{\sqrt{1-x^2}} \, dx$
 $y = \sin^{-1} x \cdot x - \frac{1}{2} \int \frac{2x}{\sqrt{1-x^2}} \, dx$
 $y = x \cdot \sin^{-1} x + \frac{1}{2} \int \frac{dt}{\sqrt{t}}$
GS: $y = x \sin^{-1} x + \sqrt{1-x^2} + C$

$$\begin{aligned}
 10) \quad e^x \tan y dx + (1 - e^x) \sec^2 y dy &= 0 \\
 e^x \tan y dx &= -(1 - e^x) \sec^2 y dy \\
 \int \frac{e^x}{(1 - e^x)} dx &= \int \frac{\sec^2 y}{\tan y} dy \\
 \log(1 - e^x) &= \log(\tan y) + C \\
 \log \left| \frac{1 - e^x}{\tan y} \right| &= C \\
 \frac{1 - e^x}{\tan y} &= C
 \end{aligned}$$

For each of the following DE's, find PS satisfying the given condition

$$11) (x^3 + x^2 + x + 1) \frac{dy}{dx} = 2x^2 + x; \quad y=1, x=0$$

$$dy = \frac{2x^2 + x}{x^2(x+1) + 1(x+1)} dx$$

$$\int dy = \int \frac{2x^2 + x}{(x^2+1)(x+1)} dx \rightarrow (1)$$

$$\frac{2x^2 + x}{(x^2+1)(x+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1}$$

$$2x^2 + x = A(x^2+1) + (Bx+C)(x+1)$$

$$x=-1: \quad 1 = A(2) \Rightarrow A = \frac{1}{2}$$

$$\text{coeff } x^2: \quad 2 = A + B \Rightarrow B = \frac{3}{2}$$

$$\text{coeff } x: \quad 1 = B + C \Rightarrow C = -\frac{1}{2}$$

$$(1) \Rightarrow \int dy = \frac{1}{2} \int \frac{dx}{x+1} + \frac{1}{2} \int \frac{3x-1}{x^2+1} dx$$

$$y = \frac{1}{2} \log|x+1| + \frac{3}{4} \int \frac{2x}{x^2+1} dx - \frac{1}{2} \int \frac{1}{x^2+1} dx$$

$$\text{GS: } y = \frac{1}{2} \log|x+1| + \frac{3}{4} \log(x^2+1) - \frac{1}{2} \tan^{-1} x + C$$

$$x=0, y=1$$

$$1 = 0 + 0 - 0 + C \Rightarrow C = 1$$

$$\text{PS: } y = \frac{1}{2} \log|x+1| + \frac{3}{4} \log(x^2+1) - \frac{1}{2} \tan^{-1} x + 1$$

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12] $x(x^2-1) \frac{dy}{dx} = 1$; $y=0, x=2$

$$\frac{dy}{dx} = \frac{1}{x(x^2-1)} = \frac{1}{x(x-1)(x+1)}$$

$$\int dy = \int \frac{-1}{x} dx + \frac{1}{2} \int \frac{1}{x-1} dx + \frac{1}{2} \int \frac{1}{x+1} dx$$

$$y = -\log|x| + \frac{1}{2} \log|x-1| + \frac{1}{2} \log|x+1| + C$$

$$y = \log \left| \frac{(x+1)(x-1)}{x} \right| + C$$

GS: $y = \log \left| \frac{x^2-1}{x} \right| + C$

$x=2, y=0$ $0 = \log \left| \frac{4-1}{2} \right| + C$

$$C = \log \left(\frac{2}{\sqrt{3}} \right)$$

PS: $y = \log \left| \frac{x^2-1}{x} \right| + \log \left(\frac{2}{\sqrt{3}} \right)$

13] $\cos \frac{dy}{dx} = A$; $y=2, x=0$

$$\frac{dy}{dx} = \cos^{-1} A$$

$$\int dy = \int \cos^{-1} A dx$$

GS: $y = x \cdot \cos^{-1} A + C$

$x=0, y=2$ $2 = 0 + C \Rightarrow C = 2$

PS: $y = x \cos^{-1} A + 2$

14] $\frac{dy}{dx} = y \tan x$; $y=1, x=0$

$$\int \frac{dy}{y} = \int \tan x dx$$

$$\log y = \sec^2 x + C$$

$$\log y = \log(\sec^2 x) + \log C$$

PS: $y = \sec^2 x \cdot C$

$x=0, y=1$
 $1 = \sec^2 0 \cdot C$
 $C = 1$

PS: $y = \sec^2 x$

15] Find the eq. of a curve passing through origin whose DE is $y' = e^x \sin x$

Sol.] Gn: $\frac{dy}{dx} = e^x \sin x$, pt = (0,0)

$$\int dy = \int e^x \sin x dx \rightarrow (1)$$

$$I = \int e^x \sin x dx$$

$$= -e^x \cos x + \int \cos x e^x dx$$

$$I = -e^x \cos x + \int e^x \sin x - \int \sin x e^x dx$$

$$2I = e^x (\sin x - \cos x)$$

$$I = \frac{e^x (\sin x - \cos x)}{2}$$

$$(1) \Rightarrow y = \frac{e^x (\sin x - \cos x)}{2} + C$$

$$x=0, y=0 \quad 0 = \frac{e^0 (\sin 0 - \cos 0)}{2} + C$$

$$0 = \frac{1}{2}(0 - 1) + C$$

$$C = \frac{1}{2}$$

$$\therefore \text{Curve : } y = \frac{e^x (\sin x - \cos x)}{2} + \frac{1}{2}$$

$$\Rightarrow 2y = e^x (\sin x - \cos x) + 1$$

16] For the DE, $xy \frac{dy}{dx} = (x+2)(y+2)$, Find the solution curve

passing through (1, -1)

Sol.] Gn: $xy \frac{dy}{dx} = (x+2)(y+2)$, pt = (1, -1)

$$\int \frac{y dy}{y+2} = \int \frac{(x+2) dx}{x}$$

$$= \int \frac{(y+2)-2}{y+2} dy = \int \left(1 + \frac{2}{x}\right) dx$$

$$x \log x = \frac{1}{2} y^2 + 2 \log x + C$$

$$\int 1 - \frac{2}{y+2} dy = \int 1 + \frac{2}{x} dx$$

$$y - 2 \log|y+2| = x + 2 \log|x| + C$$

$$x=1, y=-1$$

$$-1 - 2 \log(1) = 1 + 2 \log(1) + C$$

$$C = -2$$

$$\therefore \text{Curve : } y - 2 \log|y+2| = x + 2 \log|x| - 2$$

$$y = x + \log(x(y+2))^2 - 2$$

23] The GS of DE $\frac{dy}{dx} = e^{x+y}$ is

$$(a) e^x + e^{-y} = C$$

$$(c) e^{-x} + e^y = C$$

$$(b) e^x + e^y = C$$

$$(d) e^x + e^{-y} + C$$

$$\frac{dy}{dx} = e^{x+y}$$

$$\frac{dy}{dx} = e^x e^y$$

$$\int e^{-y} dy = \int e^x dx$$

$$-e^{-y} = e^x + C$$

$$C = e^x + e^{-y}$$

imp

EXERCISE 9.6

Find GS for the following DEs

1] $\frac{dy}{dx} + 2y = \sin x$

Form: $\frac{dy}{dx} + Py = Q$; $P=2$, $Q=\sin x$

$$IF = e^{\int P dx} = e^{\int 2 dx} = e^{2x}$$

$$GS: y(IF) = \int Q(IF) dx$$

$$y e^{2x} = \int \sin x \cdot e^{2x} dx \rightarrow (1)$$

$$\begin{aligned}
 I &= \int e^{2x} \sin x \, dx \\
 &= -e^{2x} \cos x + \int \cos x \cdot 2e^{2x} \, dx \\
 &= -e^{2x} \cos x + 2 \left[e^{2x} \sin x - \int \sin x \cdot e^{2x} \cdot 2 \, dx \right] \\
 &= -e^{2x} \cos x + 2e^{2x} \sin x - 4 \int \sin x \cdot e^{2x} \, dx
 \end{aligned}$$

$$5I = e^{2x} (2 \sin x - \cos x) - 4I$$

$$\textcircled{2} I = \frac{e^{2x}}{5} (2 \sin x - \cos x)$$

$$\textcircled{1} \Rightarrow y e^{\frac{2x}{5}} = \frac{e^{2x}}{5} (2 \sin x - \cos x) + C$$

$$2] \frac{dy}{dx} + 3y = e^{-2x}$$

$$\text{Form: } \frac{dy}{dx} + Py = Q; \quad P = 3, \quad Q = e^{-2x}$$

$$IF = e^{\int P \, dx} = e^{\int 3 \, dx} = e^{3x}$$

$$GS: y(IF) = \int Q(IF) \, dx$$

$$y e^{3x} = \int e^{-2x} e^{3x} \, dx = \int e^x \, dx = \frac{e^x}{1}$$

$$y e^{3x} = e^x + C$$

$$3] \frac{dy}{dx} + \frac{y}{x} = x^2$$

$$\text{Form: } \frac{dy}{dx} + Py = Q; \quad P = \frac{1}{x}, \quad Q = x^2$$

$$IF: e^{\int P \, dx} = e^{\int \frac{1}{x} \, dx} = e^{\log x} = x$$

$$GS: y(IF) = \int Q(IF) \, dx$$

$$y x = \int x^2 \cdot x \, dx = \int x^3 \, dx = \frac{x^4}{4} + C$$

$$xy = \frac{x^4}{4} + C$$

$$4] \frac{dy}{dx} + \sec x \cdot y = \tan x$$

$$\text{Form: } \frac{dy}{dx} + Py = Q; \quad P = \sec x, \quad Q = \tan x$$

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$$IF = e^{\int P dx} = e^{\int \sec x dx} = e^{\log(\sec x + \tan x)} = \sec x + \tan x$$

$$G.S : y(IF) = \int Q(IF) dx$$

$$y(\sec x + \tan x) = \int \tan x (\sec x + \tan x) dx$$

$$y(\sec x + \tan x) = \int \tan x \sec x + \tan^2 x dx$$

$$y(\sec x + \tan x) = \int (\sec x \tan x + \sec^2 x - 1) dx$$

$$y(\sec x + \tan x) = \sec x + \tan x - x + C$$

5) $\cos^2 x \frac{dy}{dx} + y = \tan x$ ÷ by $\cos^2 x$

$$\frac{dy}{dx} + \sec^2 x y = \tan x \sec^2 x$$

Form : $\frac{dy}{dx} + Py = Q$; $P = \sec^2 x$, $Q = \tan x \sec^2 x$

$$IF : e^{\int P dx} = e^{\int \sec^2 x dx} = e^{\tan x} = e^{\tan x}$$

$$G.S : y e^{\tan x} = \int \tan x \sec^2 x e^{\tan x} dx$$

$$y e^{\tan x} = \int t e^t dt \quad \left| \begin{array}{l} \tan x = t \\ \sec^2 x dx = dt \end{array} \right.$$

$$= t e^t - \int e^t dt$$

$$y e^{\tan x} = \tan x \cdot e^{\tan x} - e^{\tan x} + C$$

÷ by $e^{\tan x}$

$$y = \tan x - 1 + C e^{-\tan x}$$

6) $x \frac{dy}{dx} + 2y = x^2 \log x$ ÷ by x

$$\frac{dy}{dx} + \frac{2}{x} y = x \log x$$

Form : $\frac{dy}{dx} + Py = Q$; $P = \frac{2}{x}$, $Q = x \log x$

$$IF = e^{\int P dx} = e^{\int \frac{2}{x} dx} = e^{2 \log x} = x^2$$

$$G.S : y \cdot x^2 = \int x(\log x) x^2 dx = \int x^3 \log x dx$$

$$x^2 y = \log x \cdot \frac{x^4}{4} - \int \frac{x^4}{4} \cdot \frac{1}{x} dx$$

$$x^2 y = \frac{x^4}{4} \log x - \frac{1}{16} x^4 + C$$

$$7] \quad x \log x \frac{dy}{dx} + y = \frac{2}{x} \log x$$

$$\frac{dy}{dx} + \frac{1}{x \log x} y = \frac{2}{x^2}$$

$$P = \frac{1}{x \log x}; Q = \frac{2}{x^2}$$

$$IF = e^{\int P dx} = e^{\int \frac{1}{x \log x} dx} = e^{\log(\log x)} = \log x$$

$$GS: y(IF) = \int Q(IF) dx$$

$$y \log x = \int \frac{2}{x^2} \log x dx$$

$$= \log x \left(-\frac{2}{x} \right) + \int \frac{2}{x} \cdot \frac{1}{x} dx$$

$$y \log x = -\frac{2 \log x}{x} - \frac{2}{x} + C$$

$$y \log x + \frac{2 \log x}{x} + \frac{2}{x} = C$$

$$8] \quad (1+x^2) dy + 2xy dx = \cot x dx$$

$$\frac{dy}{dx} + \frac{2x}{1+x^2} y = \frac{\cot x}{1+x^2} \quad \div \text{ by } (1+x^2) dx$$

$$\text{Form: } \frac{dy}{dx} + Py = Q \Rightarrow P = \frac{2x}{1+x^2}; Q = \frac{\cot x}{1+x^2}$$

$$IF = e^{\int P dx} = e^{\int \frac{2x}{1+x^2} dx} = e^{\log(1+x^2)} = 1+x^2$$

$$GS: y(IF) = \int Q(IF) dx$$

$$y(1+x^2) = \int \frac{\cot x}{1+x^2} (1+x^2) dx = \int \cot x dx = \log|\sin x| + C$$

$$y(1+x^2) = \log|\sin x| + C$$

$$9] \quad x \frac{dy}{dx} + y - x + xy \cot x = 0$$

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$$\frac{dy}{dx} + \left(\frac{1}{x} + \cot x\right)y = 1$$

Form : $P = \frac{1}{x} + \cot x$; $Q = 1$

$$IF : e^{\int P dx} = e^{\int \frac{1}{x} + \cot x} = e^{\log x + \log \sin x} = x \sin x$$

$$GS : y(IF) = \int Q(IF) dx$$

$$y \cdot x \sin x = \int 1 \cdot x \sin x dx$$

$$xy \sin x = x(1 - \cos x) + \int \cos x dx$$

$$xy \sin x + x \cos x = \sin x + C$$

*** 10) $(x+y)\frac{dy}{dx} = 1$

$$x \frac{dx}{dy} + y \frac{dy}{dy} = 1$$

$$x + y = \frac{dx}{dy}$$

$$\Rightarrow \frac{dx}{dy} - x = y$$

Form : $\frac{dx}{dy} + Px = Q$; $P = -1$, $Q = y$

$$IF : e^{\int P dy} = e^{\int -1 dy} = e^{-y}$$

$$GS : x(IF) = \int Q(IF) dy$$

$$xe^{-y} = \int y \cdot e^{-y} dy$$

$$= y(-e^{-y}) + \int e^{-y} dy$$

$$xe^{-y} + e^{-y} + e^{-y} = C$$

$$\div \text{by } e^{-y}$$

$$x + y + 1 = \frac{C}{e^{-y}}$$

*** 11) $y dx + (x - y^2) dy = 0$

$$y dx = (y^2 - x) dy$$

$$\frac{dx}{dy} + \frac{x}{y} = y$$

Form : $\frac{dx}{dy} + Px = Q \Rightarrow P = \frac{1}{y}$; $Q = y$

$$IF = e^{\int P dy} = e^{\int \frac{1}{y} dy} = e^{\log y} = y$$

$$QS: x(IF) = \int Q(IF) dy$$

$$xy = \int y^2 dy$$

$$xy = \frac{y^3}{3} + C$$

$$^{**} 12] (x + 3y^2) \frac{dy}{dx} = y$$

$$(x + 3y^2) = y \frac{dx}{dy}$$

$$\frac{dx}{dy} = \frac{x}{y} + 3y$$

$$\frac{dx}{dy} - x\left(\frac{1}{y}\right) = 3y$$

$$\text{Form: } P = -\frac{1}{y}; Q = 3y$$

$$IF = e^{\int P dy} = e^{\int -\frac{1}{y} dy} = e^{-\log y} = \frac{1}{y}$$

$$QS: x(IF) = \int Q(IF) dy$$

$$x\left(\frac{1}{y}\right) = \int 3y\left(\frac{1}{y}\right) dy$$

$$\frac{x}{y} = 3y + C$$

For each of the following find PS, satisfying the condition.

$$13] \frac{dy}{dx} + 2y \tan x = \sin x; y=0, x=\frac{\pi}{3}$$

$$\text{Form: } \frac{dy}{dx} + Py = Q \Rightarrow P = 2 \tan x; Q = \sin x$$

$$IF: e^{\int P dx} = e^{\int 2 \tan x dx} = e^{2 \log \sec x} = e^{\log \sec^2 x} = \sec^2 x$$

$$QS: y(IF) = \int Q(IF) dx$$

$$y \sec^2 x = \int \sin x \cdot \sec^2 x dx = \int \frac{\sin x}{\cos x} \sec x dx$$

$$y \sec^2 x = \int \tan x \sec x dx$$

$$QS: y \sec^2 x = \sec x + C$$

$$x = \frac{\pi}{3}, y = 0$$

$$0 = 2 + C \Rightarrow C = -2$$

$$\Rightarrow \text{PS: } y \sec^2 x = \sec x - 2$$

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14] $(1+x^2) \frac{dy}{dx} + 2xy = \frac{1}{1+x^2}$; $y=0, x=1$

$\div$ by $(1+x^2)$

$$\frac{dy}{dx} + y \left(\frac{2x}{1+x^2} \right) = \frac{1}{(1+x^2)^2}$$

Form : $P = \frac{2x}{1+x^2}$, $Q = \frac{1}{(1+x^2)^2}$

IF : $e^{\int \frac{2x}{1+x^2} dx} = e^{\log(1+x^2)} = 1+x^2$

GS : $y(1+x^2) = \int \frac{1}{(1+x^2)^2} (1+x^2) dx$

$$(1+x^2)y = \int \frac{1}{1+x^2} dx$$

GS : $(1+x^2)y = \tan^{-1}x + C$

$x=1, y=0$

$$0 = \frac{\pi}{4} + C \Rightarrow C = -\frac{\pi}{4}$$

PS : $(1+x^2)y = \tan^{-1}x - \frac{\pi}{4}$

** 15] $\frac{dy}{dx} - 3y \cot x = \sin 2x$; $y=2, x=\frac{\pi}{2}$

Form : $P = -3 \cot x$, $Q = \sin 2x$

IF : $e^{\int -3 \cot x dx} = e^{-3 \log(\sin x)} = e^{\log(\operatorname{cosec}^3 x)} = \operatorname{cosec}^3 x$

GS : $y \cdot \operatorname{cosec}^3 x = \int \sin 2x \cos x \cdot \operatorname{cosec}^3 x dx$

$$= \int \frac{2 \sin x \cos x}{\sin x \cdot \sin^2 x} dx = \int 2 \operatorname{cosec} x \cot x dx$$

GS : $y \operatorname{cosec}^3 x = -2 \operatorname{cosec} x + C$

$y=2, x=\frac{\pi}{2}$

$$2(1) = -2(1) + C \Rightarrow C = 4$$

$\therefore$ PS : $y \operatorname{cosec}^3 x + 2 \operatorname{cosec} x = 4$

18] The IF of the DE: $x \frac{dy}{dx} - y = 2x^2$ is $\frac{1}{x}$

$$\div \text{ by } x \quad \frac{dy}{dx} - \left(\frac{1}{x}\right)y = 2x$$

$$P = -\frac{1}{x}, Q = 2x$$

$$\text{IF: } e^{\int -\frac{1}{x} dx} = e^{-\log(x)} = \frac{1}{x}$$

19] IF of the DE $(1-y^2) \frac{dx}{dy} + yx = ay$

$$\frac{dx}{dy} + x \left(\frac{y}{1-y^2}\right) = \frac{ay}{1-y^2}$$

$$P = \frac{y}{1-y^2}; Q = \frac{ay}{1-y^2}$$

$$\text{IF: } e^{\int P dy} = e^{\int \frac{y}{1-y^2} dy} = e^{-\frac{1}{2} \log(1-y^2)} = \frac{1}{\sqrt{1-y^2}}$$

16] Find the eq. of a curve passing through origin given that slope of the tangent to the curve at any point (x, y) is equal to the sum of the coordinates of the point.

Given: pt = $(0, 0)$

m of T = $\frac{dy}{dx}$ = sum of coordinates

$$\frac{dy}{dx} = x + y$$

$$\frac{dy}{dx} - y = x$$

$$\text{Form: } P = -1, Q = x$$

$$\text{IF: } e^{\int -1 dx} = e^{-x}$$

$$\text{Ans: } ye^{-x} = \int x \cdot e^{-x} dx$$

$$= -xe^{-x} + \int e^{-x}(-1) dx = -xe^{-x} - e^{-x} + c$$

$$ye^{-x} = -xe^{-x} - e^{-x} + ce^x \quad \div \text{ by } e^{-x}$$

$$y + x = -1 + ce^x$$

$$\text{At pt } (0, 0) \quad 0 + 0 = -1 + ce^0 \Rightarrow c = 1$$

$$\therefore \text{The curve is } \boxed{x + y + 1 = e^x}$$

- 17] Find the eq. of the curve passing through $(0, 2)$, given that sum of coordinates of any point on the curve exceeds the magnitude of m of T to the curve at that point by 5.

Gn: $pt = (0, 2)$

$m \text{ of } T + 5 = \text{sum of coordinates}$

$$\frac{dy}{dx} + 5 = x + y$$

$$\frac{dy}{dx} - y = x - 5$$

Form: $P = -1, Q = x - 5$

$$I.F. = e^{\int -1 dx} = e^{-x}$$

$$Gn: ye^{-x} = \int (x-5)e^{-x} dx = \int (xe^{-x} - 5e^{-x}) dx$$

$$= [-xe^{-x} + \int e^{-x} dx] - \int 5e^{-x}$$

$$ye^{-x} = -xe^{-x} - e^{-x} + 5e^{-x} + C$$

$$y = -x - 1 + 5 + ce^x$$

$pt = (0, 2)$

$$2 = 0 - 1 + 5 + ce^0 \Rightarrow C = -2$$

$\therefore$ The curve is $y = 4 - x - 2e^x$

Homogeneous DE:

EXERCISE 9.5

NOTE: * If the variables can't be separated, then check for homogeneity.

* If $F(\lambda x, \lambda y) = \lambda^n F(x, y)$

then the given DE is homogeneous, λ representing the degree of components of DE.

** Eg 15] ST the DE $(x-y)\frac{dy}{dx} = x+2y$ is homogeneous and hence solve it.

Gn: $(x-y)\frac{dy}{dx} = x+2y$

$$\frac{dy}{dx} = \frac{x+2y}{x-y}$$

Degree = 1

Degree = 1

$$F(x, y) = \frac{x+2y}{x-y}$$

$$F(\lambda x, \lambda y) = \frac{\lambda(x+2y)}{\lambda(x-y)} = \lambda^0 F(x, y)$$

$\Rightarrow$ Gen. DE is homogeneous.

$$\frac{dy}{dx} = \frac{x+2y}{x-y} \quad \left| \begin{array}{l} \text{put } y = vx \Rightarrow v = \frac{y}{x} \\ \frac{dy}{dx} = v + x \frac{dv}{dx} \end{array} \right.$$

$$v + x \frac{dv}{dx} = \frac{x+2vx}{x-vx}$$

$$x \frac{dv}{dx} = \frac{1+2v}{1-v} - v = \frac{1+2v-v+v^2}{1-v}$$

$$x \frac{dv}{dx} = \frac{v^2+v+1}{1-v}$$

$$\int \frac{1-v}{v^2+v+1} dv = \int \frac{dx}{x}$$

$Nu = A \frac{d}{dx}(de) + B$
 $1-v = A(2v+1) + B$
 coeff v : $-1 = 2A \Rightarrow A = -\frac{1}{2}$
 constants: $1 = A+B \Rightarrow B = \frac{3}{2}$

$$\int \frac{1-v}{v^2+v+1} dv = \frac{1}{2} \int \frac{-1(2v+1) + 3}{v^2+v+1} dv = -\frac{1}{2} \int \frac{2v+1}{v^2+v+1} dv + \frac{3}{2} \int \frac{dv}{v^2+v+1}$$

$$v^2+v+1 = t \quad \left| \begin{array}{l} v^2+v+1 \\ (2v+1)dv = dt \end{array} \right. \quad \left(v + \frac{1}{2} \right)^2 + 1 - \frac{1}{4} = \left(v + \frac{1}{2} \right)^2 + \left(\frac{\sqrt{3}}{2} \right)^2$$

$$\int \frac{1-v}{v^2+v+1} dv = -\frac{1}{2} \int \frac{dt}{t} + \frac{3}{2} \int \frac{dt}{\left(v + \frac{1}{2} \right)^2 + \left(\frac{\sqrt{3}}{2} \right)^2}$$

$$= -\frac{1}{2} \log(v^2+v+1) + \frac{3}{2} \cdot \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{v + \frac{1}{2}}{\frac{\sqrt{3}}{2}} \right)$$

$$= -\log \sqrt{v^2+v+1} + \sqrt{3} \tan^{-1} \left(\frac{2v+1}{\sqrt{3}} \right)$$

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$$-\log \sqrt{v^2+v+1} + \sqrt{3} \tan^{-1} \left(\frac{2v+1}{\sqrt{3}} \right) = \log x + C = \log x + C$$

$$\sqrt{3} \tan^{-1} \left(\frac{2\frac{y}{x}+1}{\sqrt{3}} \right) = \log \left(x \sqrt{\frac{y^2}{x^2} + \frac{y}{x} + 1} \right) + C$$

1.] $(x^2 + xy) dy = (x^2 + y^2) dx$

$$\frac{dy}{dx} = \frac{x^2 + y^2}{x^2 + xy} = F(x, y)$$

$$F(\lambda x, \lambda y) = \frac{\lambda(x^2 + y^2)}{\lambda(x^2 + xy)} = \frac{x^2 + y^2}{x^2 + xy} = F(x, y)$$

$\Rightarrow$ G.D.E is homogeneous

$$\frac{dy}{dx} = \frac{x^2 + y^2}{x^2 + xy}$$

$$y = vx \quad \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = \frac{x^2 + v^2 x^2}{x^2 + x^2 v} = \frac{1 + v^2}{1 + v}$$

$$x \frac{dv}{dx} = \frac{1 + v^2}{1 + v} - v = \frac{1 + v^2 - v - v^2}{1 + v} = \frac{1 - v}{1 + v}$$

$$x \frac{dv}{dx} = \frac{1 - v}{1 + v}$$

$$\int \frac{(1+v)^2 dv}{(1-v)^2} = \int \frac{dx}{x}$$

$$\int \frac{1 + 2v + v^2}{1 - 2v + v^2} dv = \log x + C$$

$$-\int \frac{1+v}{v-1} dv = \int \frac{dx}{x}$$

$$-\int \frac{1+(v-1)+1}{v-1} dv = \log x + C$$

$$-\int \left(1 + \frac{2}{v-1} \right) dv = \log x + C$$

$$-[v + 2 \log(v-1)] = \log x + C$$

$$\frac{y}{x} + 2 \log \left(\frac{y}{x} - 1 \right) + \log x = C$$

2] $\frac{dy}{dx} = \frac{x+y}{x}$ $\left| \begin{array}{l} \text{Let } y = vx \\ \frac{dy}{dx} = v + x \frac{dv}{dx} \end{array} \right.$

$$v + x \frac{dv}{dx} = \frac{x+vx}{x} = 1+v$$

$$x \frac{dv}{dx} = 1+v - v = 1$$

$$\int dv = \int \frac{dx}{x}$$

$$v = \log x + C$$

$$\frac{y}{x} = \log x + C$$

3] $(x-y)dy - (x+y)dx = 0$

$$\frac{dy}{dx} = \frac{x+y}{x-y} = F(x, y)$$

$$F(\lambda x, \lambda y) = \frac{\lambda(x+y)}{\lambda(x-y)} = F(x, y)$$

$\Rightarrow$ Given DE is homogeneous.

$$\frac{dy}{dx} = \frac{x+y}{x-y}$$

$$y = vx$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = \frac{x+vx}{x-vx}$$

$$v + x \frac{dv}{dx} = \frac{1+v}{1-v}$$

$$x \frac{dv}{dx} = \frac{1+v}{1-v} - v = \frac{1+v - v(1-v)}{1-v} = \frac{1+v - v + v^2}{1-v} = \frac{1+v^2}{1-v}$$

$$x \frac{dv}{dx} = \frac{1+v^2}{1-v}$$

$$\int \frac{1-v}{1+v^2} dv = \int \frac{dx}{x}$$

$$\int \frac{1}{1+v^2} dv - \frac{1}{2} \int \frac{2v}{1+v^2} dv = \log x + C$$

$$\tan^{-1} v - \frac{1}{2} \log(1+v^2) = \log x + C$$

$$\tan^{-1} \frac{y}{x} = \frac{1}{2} \log \left(1 + \frac{y^2}{x^2} \right) + \log x + C$$

$$\tan^{-1} \frac{y}{x} = \log(\sqrt{x^2+y^2}) + C$$

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8] $x \frac{dy}{dx} - y + x \sin\left(\frac{y}{x}\right) = 0$

$\frac{dy}{dx} - \frac{y}{x} + \sin\left(\frac{y}{x}\right) = 0$

$\frac{dy}{dx} = \frac{y}{x} - \sin\left(\frac{y}{x}\right)$

$v + x \frac{dv}{dx} = \frac{vx}{x} - \sin\left(\frac{vx}{x}\right)$

$x + x \frac{dv}{dx} = x - \sin v$

$x \frac{dv}{dx} = -\sin v$

$\int \frac{dv}{\sin v} = -\int \frac{dx}{x}$

$\int \operatorname{cosec} v \, dv = -\int \frac{dx}{x}$

$\log |\operatorname{cosec} v - \cot v| = -\log x + \log C$

$(\operatorname{cosec} v - \cot v) x = C$

$\left(\operatorname{cosec} \frac{y}{x} - \cot \frac{y}{x}\right) x = C$

$\left(\frac{1}{\sin \frac{y}{x}} - \frac{\cos \frac{y}{x}}{\sin \frac{y}{x}}\right) x = C$

$(1 - \cos \frac{y}{x}) x = C \sin\left(\frac{y}{x}\right)$

9] $y dx + x \log\left(\frac{y}{x}\right) dy - 2x dy = 0$

$y dx + dy [x \log\left(\frac{y}{x}\right) - 2x] = 0$

$\frac{dy}{dx} = \frac{-y}{x \log\left(\frac{y}{x}\right) - 2x}$

$v + x \frac{dv}{dx} = \frac{v}{2 - \log v}$

$x \frac{dv}{dx} = \frac{v}{2 - \log v} - v$

$x \frac{dv}{dx} = \frac{v - 2v + v \log v}{2 - \log v} = \frac{v(\log v - 1)}{2 - \log v}$

$$\int \frac{2 - \log v}{v(\log v - 1)} dv = \int \frac{dx}{x}$$

$$\int \frac{1 - (\log v - 1)}{v(\log v - 1)} dv = \int \frac{dx}{x}$$

$$\int \left(\frac{1}{v(\log v - 1)} - \frac{1}{v} \right) dv = \int \frac{dx}{x}$$

$$\log |\log v - 1| - \log |v| = \log |x| + \log C$$

$$\log \frac{\log v - 1}{v} = \log x + \log C$$

$$\frac{\log \frac{y}{x} - 1}{\frac{y}{x}} = xC$$

$$\left(\log \frac{y}{x} - 1 \right) \frac{x}{y} = xC$$

$$\log \frac{y}{x} - 1 = yC$$

14] $\frac{dy}{dx} - \frac{y}{x} + \operatorname{cosec}\left(\frac{y}{x}\right) = 0$; $y=0, x=1$

$$\frac{dy}{dx} = \frac{y}{x} - \operatorname{cosec}\left(\frac{y}{x}\right)$$

$$v + x \frac{dv}{dx} = \frac{vx}{x} - \operatorname{cosec}\left(\frac{vx}{x}\right)$$

$$x \frac{dv}{dx} = -\operatorname{cosec} v$$

$$\int \frac{dv}{\operatorname{cosec} v} = \int -\frac{dx}{x}$$

$$-\cos v = -\log x + C$$

GS: $\log x - \cos \frac{y}{x} = C$

$x=1, y=0$

$$\log 1 - \cos 0 = C$$

$$-1 = C$$

PS: $\log x - \cos \frac{y}{x} = -1$

$$\log x + \log e = \cos \frac{y}{x}$$

$$\log |xe| = \cos \frac{y}{x}$$

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15] $2xy + y^2 - 2x^2 \frac{dy}{dx} = 0$; $y=2, x=1$

$$2xy + y^2 = 2x^2 \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{2xy + y^2}{2x^2}$$

$$v + x \frac{dv}{dx} = \frac{2xvx + x^2v^2}{2x^2} = \frac{x^2(2v + v^2)}{2x^2}$$

$$v + x \frac{dv}{dx} = v + \frac{v^2}{2}$$

$$\int \frac{2}{v^2} dv = \int \frac{dx}{x}$$

$$-\frac{2}{v} = \log x + C$$

$$-\frac{2}{\frac{y}{x}} = \log x + C$$

G.S: $-\frac{2x}{y} = \log x + C$

$y=2, x=1$: $-1 = C$

P.S: $\log x + \frac{2x}{y} = +1$

$$-\frac{2x}{y} = \log x - 1$$

$$y = \frac{-2x}{\log x - 1}$$

P.S: $y = \frac{2x}{1 - \log x}$

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6] $x dy - y dx = \sqrt{x^2 + y^2} dx$

$$x dy = (\sqrt{x^2 + y^2} + y) dx$$

$$\frac{dy}{dx} = \frac{\sqrt{x^2 + y^2} + y}{x}$$

Let $y = vx$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = \frac{\sqrt{x^2 + v^2 x^2} + vx}{x} = \sqrt{1 + v^2} + v$$

$$x \frac{dv}{dx} = \sqrt{1 + v^2}$$

$$\int \frac{dv}{\sqrt{1 + v^2}} = \int \frac{dx}{x}$$

$$\log |v + \sqrt{1 + v^2}| = \log x + \log c$$

$$\frac{y}{x} + \frac{\sqrt{x^2 + y^2}}{x} = Cx$$

$$y + \sqrt{x^2 + y^2} = Cx^2$$

10] $(1 + e^{xy}) dx + e^{xy} \left(1 - \frac{x}{y}\right) dy = 0$

$\left(\frac{x}{y}\right)$ is dominative

$$(1 + e^{xy}) dx = e^{xy} \left(\frac{x}{y} - 1\right) dy$$

$$\frac{dx}{dx} = \frac{e^{xy} \left(\frac{x}{y} - 1\right)}{1 + e^{xy}}$$

$$\frac{dx}{dy} = \frac{e^{xy} \left(\frac{x}{y} - 1\right)}{(1 + e^{xy})}$$

put $x = vy$

$$\frac{dx}{dy} = v + y \frac{dv}{dy}$$

$$v + y \frac{dv}{dy} = \frac{e^v (v - 1)}{(1 + e^v)}$$

$$y \frac{dv}{dy} = \frac{ve^v - e^v - v - ve^v}{1 + e^v}$$

$$\int \frac{1 + e^v}{v + e^v} dv = - \int \frac{dy}{y}$$

$$\log |v + e^v| + \log y = \log c$$

$$\left(\frac{x}{y} + e^{xy}\right) \cdot y = c$$

$$11) (x+y)dy + (x-y)dx = 0 \quad \text{Let } y = vx, \quad x = \log - \log x \quad (1)$$

$$\frac{dy}{dx} = \frac{y-x}{x+y}$$

$$v + x \frac{dv}{dx} = \frac{vx-x}{-x+vx}$$

$$x \frac{dv}{dx} = \left(\frac{v-1}{1+v} \right) - v$$

$$x \frac{dv}{dx} = \frac{v-1-v-v^2}{1+v}$$

$$x \frac{dv}{dx} = \frac{-1-v^2}{1+v}$$

$$\int \frac{1+v}{-1-v^2} dv = \int \frac{dx}{x}$$

$$-\int \frac{dv}{(1+v^2)} - \frac{1}{2} \int \frac{2v}{(1+v^2)} = \int \frac{dx}{x}$$

$$-\tan^{-1} v - \frac{1}{2} \log(1+v^2) = \log x + C$$

$$\log x + \tan^{-1} v + \log \sqrt{1+v^2} = C$$

$$\log x + \tan^{-1} \left(\frac{y}{x} \right) + \log \sqrt{\frac{x^2+y^2}{x^2}} = C$$

$$\text{G.S.: } \log \sqrt{x^2+y^2} + \tan^{-1} \left(\frac{y}{x} \right) = C$$

$$x=1, y=1$$

$$\log \sqrt{2} + \frac{\pi}{4} = C$$

$$\text{P.S.: } \log \sqrt{x^2+y^2} + \tan^{-1} \frac{y}{x} = \frac{\pi}{4} + \log \sqrt{2}$$

$$12) x^2 dy + (xy+y^2) dx = 0 \quad (1-v)^2 \quad x=1, y=1$$

$$\frac{dy}{dx} = -\frac{(xy+y^2)}{x^2}$$

$$v + x \frac{dv}{dx} = -\frac{(x^2v + x^2v^2)}{x^2}$$

$$v + x \frac{dv}{dx} = -(v+v^2)$$

$$x \frac{dv}{dx} = -v-v^2-v$$

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$$\int \frac{dv}{-2v-v^2} = \int \frac{dx}{x}$$

$$\int \frac{dv}{2v+v^2} = -\int \frac{dx}{x}$$

$$\int \frac{dv}{(v+1)^2-1} = -\log x + \log C$$

$$\frac{1}{2} \log \left| \frac{v+1-1}{v+1+1} \right| + \log x = \log C \quad \times 2 \text{ by } 2$$

$$\log x^2 \left(\frac{y}{x} \right) = C^2$$

$$\left(\frac{x^2 y}{2x+y} \right) = C^2$$

GS : $x^2 y = C^2 (2x+y)$

$x=1, y=1 \Rightarrow 1 = C^2(3) \Rightarrow C = \pm \frac{1}{\sqrt{3}} \Rightarrow C^2 = \frac{1}{3}$

PS : $x^2 y = \frac{1}{3}(2x+y)$

$$3x^2 y = 2x+y$$

16] A homogeneous DE of the form $\frac{dx}{dy} = h \frac{x}{y}$ can be solved by the substitution

Ans] $x = vy$

** 17] Which of the following is homogeneous DE

(a) $(4x+6y+5)dy - (3y+2x+4)dx = 0$

(b) $(xy)dx - (x^3+y^3)dy = 0$

(c) $(x^2+2y^2)dx + 2xydy = 0$

(d) $y^2dx + (x^2-xy-y^2)dy = 0$